Prediction in Economic Analysis

The Use of Supervised Machine Learning

Inference vs Prediction · Loss and Penalty · Ridge · LASSO

Gigih Fitrianto | Week 2 | Undergraduate Economics

From Week 1 to Week 2: same question, new methods

Week 1 | What we know

- Predict 2025 poverty with 2024 indicators.
- Fit baseline, simple and multiple OLS.
- Compare MAE and RMSE on the untouched holdout.
- Inspect actual–predicted gaps.

Week 2 | What we investigate

- What loss does OLS minimize?
- What happens when we penalize coefficients?
- Can Ridge or LASSO generalize better?
- Does regularization help *this* economic dataset?

One dataset. A common evaluation design. Different objectives.

Learning objectives

1. Identify the observed **target** Y that supervises model fitting, including OLS.
2. Distinguish **coefficient precision** from **out-of-sample accuracy** under multicollinearity.
3. Explain **loss functions**, training fit, validation loss and test accuracy.
4. Interpret the role of a **penalty** and its tuning parameter λ.
5. Explain how **Ridge** and **LASSO** affect correlated predictors and coefficients.
6. Compare prediction methods in R without leaking test information.
7. Interpret forecast differences as **empirical findings**, not guaranteed ML advantages.

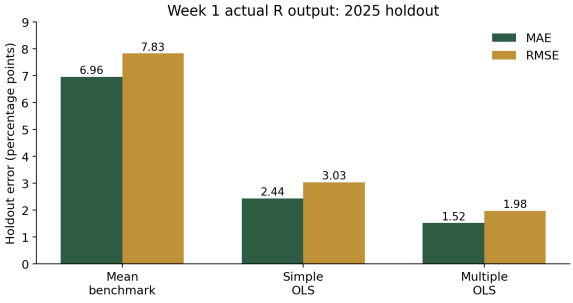
Workshop plan | 180 minutes (selected slides for live teaching)

Block	Focus	Time
A	Week 1 recap; inference vs prediction; loss functions	50 min
B	Multicollinearity, regularization, Ridge and LASSO	45 min
Break	Questions and setup	10 min
C	R practice: regularization, tuning and 2025 predictions	55 min
D	Interpretation, limitations and reflection	20 min

Our guiding question

Does controlling coefficient size improve next-year regional poverty prediction relative to OLS?

Week 1 benchmark: actual submitted R output



	MAE	RMSE
Training mean	6.957	7.827
Simple OLS	2.439	3.034
Multiple OLS	**1.523**	**1.977**

Interpretation

Six-predictor OLS was most accurate among the three Week 1 rules on the **2025 holdout**.

Actual Week 1 R results; 120 fictitious regions in 2025, errors in poverty percentage points. All data are synthetic.

The same economic prediction task

Target and predictors

y_{it} = poverty rate in region i and year t.

$X_{i,t-1}$ contains prior-year GRDP per capita, unemployment, schooling, internet access, manufacturing share and investment.

Unchanged design

- 120 fictitious regions
- 2020–2025 original panel
- 2021–2024 model-training outcomes ($n = 480$)
- 2025 holdout outcomes ($n = 120$)

Important

We compare prediction rules on *identical observations* and predict the same outcome using the same six predictors.

Econometrics and ML: where is the difference?

One important clarification

OLS is a supervised learning algorithm when fitted to observed (X, Y) pairs.

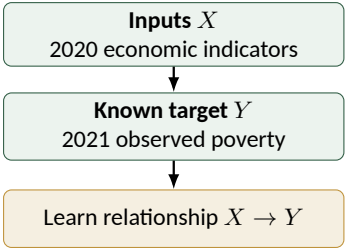
	Explanatory / inferential OLS	**Predictive OLS / regularized ML**
Outcome Y known in training	Yes	Yes
How coefficients are obtained	Minimize squared errors	Minimize loss (plus penalty for Ridge/LASSO)
Primary research goal	Interpret coefficients; test hypotheses	Generalize predictions to unseen outcomes
Evidence of success	Assumptions, uncertainty, identification	Validation and held-out prediction error

Neither loss minimization nor regularization defines all ML. The distinction concerns the learning task, evaluation design and modeling choices. Predictive association does not establish causality.

Why is it called supervised learning?

The "supervision" is the known outcome

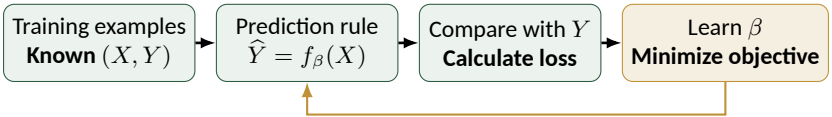
The training data contain **both** the economic indicators X and the corresponding **observed outcome** Y. The known Y provides the target against which the model's prediction is evaluated.



- X: prior-year economic indicators.
- Y: observed regional poverty rate.
- $\widehat{Y} = f_{\beta}(X)$: predicted poverty.
- The gap $Y - \widehat{Y}$ is a learning signal.

"Label" is ML terminology for the observed target: for regression, it can be a continuous number. Supervised does not mean that a person manually corrects every prediction.

Where does supervision enter model training?



OLS and Ridge/LASSO share this training logic

OLS minimizes squared errors; Ridge/LASSO minimize squared errors plus a penalty. The known Y supervises both. Testing on unseen outcomes assesses whether the learned rule generalizes.

Why does the presence of a target matter?

Supervised: predict an outcome

Question: How much poverty next year?

Training information: past indicators X *and* observed poverty Y.

Feedback: prediction loss against known Y.

Methods: OLS, Ridge, LASSO, random forest.

Unsupervised: describe structure

Question: Which regions look similar?

Training information: economic characteristics X, with no predefined target Y.

Criterion: variance explained, within-cluster similarity, etc.

Methods: PCA and clustering (Week 3).

The supervision refers to the task and observed target, not the regression formula or an intervention by the instructor.

Connecting Week 1 directly to supervised learning

1. **Train:** 2021–2024 poverty rates were observed, alongside prior-year X. OLS learned its coefficients from these labeled examples.
2. **Predict:** Given 2024 indicators, OLS produced a prediction of 2025 poverty without using the actual 2025 poverty outcomes.
3. **Evaluate:** Compare 2025 predictions to observed 2025 poverty *only after fitting*. Compute MAE and RMSE.
4. **Extend in Week 2:** Use the same training labels, replace the training objective with Ridge/LASSO penalization, and compare on the same 2025 holdout.

Key message

Week 1 was already an exercise in supervised prediction. Week 2 introduces **complexity control** and **hyperparameter validation**.

The essential question: what is a loss function?

Loss function $\ell(y, \widehat{y})$

A rule assigning a numerical cost to the gap between an observed outcome and its predicted value.

Observed outcome

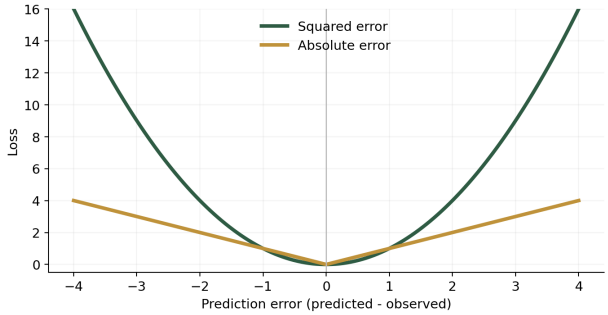
Poverty is 12% in a fictitious district.

Model prediction

The model predicts 15%: signed error $= +3$ percentage points.

How should a model penalize an error of three points?

Figure reading: squared versus absolute loss



Read the axes

- Horizontal: signed error, $\widehat{Y} - Y$.
- Vertical: cost assigned to that error.

Read the curves

- Green MSE grows quadratically.
- Gold absolute loss grows linearly.

Role in supervised ML

Known outcomes Y make errors measurable. The chosen loss determines which types of prediction mistakes the algorithm prioritizes when fitting a model.

Illustrative functions: not actual regression results for the poverty dataset.

Two familiar choices: MSE and MAE

Mean squared error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$$

Large errors receive a larger relative penalty because they are squared.

Mean absolute error (MAE)

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|$$

Measured in the original outcome units; comparatively less sensitive to large errors.

$\text{RMSE} = \sqrt{\text{MSE}}$ is also in the original units. In our example, errors are percentage points.

Mini-exercise: two prediction rules

Rule	Error 1	Error 2	Error 3
A	0	0	6
B	2	2	2

Compare MAE

A: 2; B: 2. A tie.

Compare MSE

A: 12; B: 4. B wins.

The evaluation metric expresses what we value in a forecast.

OLS: choosing intercept and slope to minimize loss

$$\widehat{y}_i = a + bx_i$$

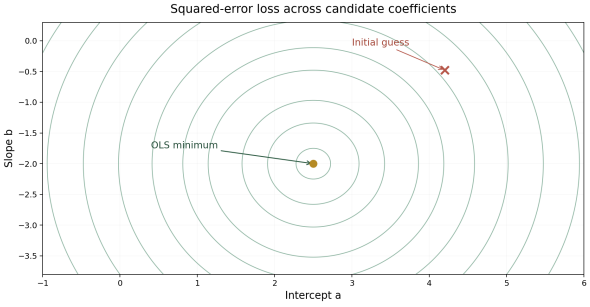
$$\min_{a,b} \frac{1}{n}\sum_{i=1}^{n}(y_i - a - bx_i)^2$$

What Week 1's Shiny sliders demonstrated

A manually chosen a and b may not minimize squared error. OLS estimates the pair with the smallest training sum of squared residuals.

OLS minimizes SSE or equivalently MSE (same minimizer for fixed n).

Figure reading: the OLS loss landscape



Horizontal axis: intercept a.

Vertical axis: slope b.

Contours: pairs (a, b) with identical training MSE.

Minimum: the best least-squares fit to known training Y.

Crosses: a non-optimal initial guess.

Role in supervised ML

The observed target creates a measurable loss surface. OLS finds its minimum; regularized regression modifies the surface by adding a coefficient penalty.

Schematic one-predictor example with simulated data; not the Week 1 coefficient estimates.

OLS estimates by optimization, not a different model family

What is learned?

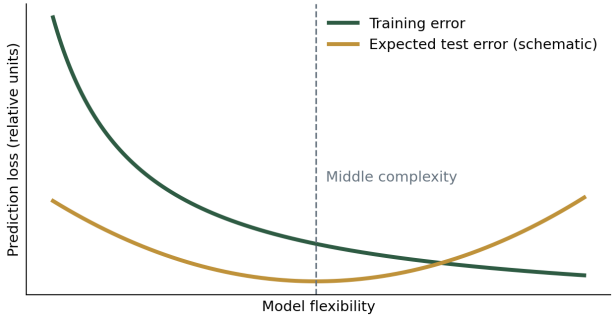
- Intercept and coefficients
- Fitted values
- A linear prediction rule

What is *not* automatic?

- Stability with correlated predictors
- Accuracy on unseen data
- Causal interpretation

Minimizing training loss is only one piece of predictive modeling.

Figure reading: training fit versus generalization



X-axis: model flexibility.

Y-axis: prediction error.

Green: training error generally falls as flexibility rises.

Gold: expected error on new data may eventually rise.

A model can fit training labels too closely.

Role in supervised ML

Training uses the known Y, while validation measures errors on different known outcomes. This distinction helps detect overfitting before final testing.

Schematic relationships, not empirically measured Week 1 learning curves.

Prediction bias and variance: a careful distinction

$$\mathbb{E}\Big[(Y - \widehat{f}(x))^2 \mid X = x\Big] = \mathsf{Bias}(\widehat{f}(x))^2 + \mathsf{Var}(\widehat{f}(x)) + \sigma^2(x)$$

- **Prediction bias:** systematic difference between mean predictions and the conditional outcome mean.
- **Variance:** how much fitted predictions change across training samples.
- **Irreducible error:** uncertainty remaining even with the ideal mean predictor.

Pointwise squared-error decomposition, with independent zero-mean outcome noise. Prediction bias differs from OLS coefficient bias and causal identification bias.

Why might standard OLS become unstable?

Predictor-related issues

- Strongly correlated economic indicators
- Many covariates relative to sample size
- Interactions and transformed variables

Statistical consequences

- Large changes in fitted coefficients
- High variability between samples
- Overfitting, even if training R^2 improves

Example

GRDP per capita, schooling and internet access may contain overlapping predictive information. With only six predictors and 480 training rows, regularization may have *limited* additional benefit.

Inference and prediction ask different questions

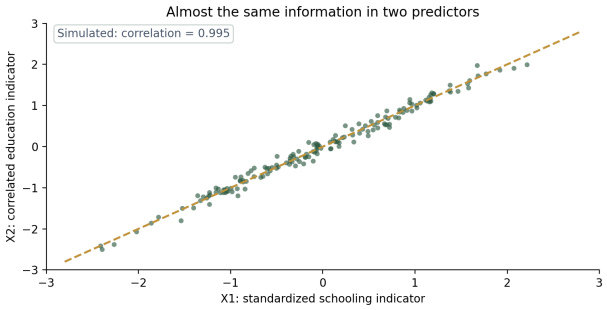
	Statistical inference	Predictive modeling
Question	What is the partial association of schooling, *holding other X fixed*?	Can last year's indicators forecast next-year poverty?
Object of interest	$\hat{\beta}_j$, standard error, confidence interval	$\widehat{Y}$, forecast errors on unseen outcomes
Multicollinearity	Less independent variation to distinguish correlated coefficients	May still provide useful combined signal
Assessment	Precision, assumptions, identification strategy	Out-of-sample MAE, RMSE, stability

Main teaching distinction

The same **OLS algorithm** can support either objective. The concern about correlated predictors depends on *what we want to learn* from the fitted model.

Predictive accuracy does not establish a causal effect. Correlated predictors are not automatically a reason to abandon OLS for forecasting.

What is multicollinearity?



Interpret the figure

- Both axes are standardized economic indicators.
- The points form a narrow diagonal band.
- One predictor provides little *additional* variation given the other.

Near versus perfect collinearity

Near collinearity weakens separate coefficient precision; *perfect* collinearity means the individual coefficients cannot be uniquely identified.

Simulated plot ($\rho \approx 0.995$), not correlations measured in the Week 1 poverty panel.

Inference: why separate coefficients become imprecise

OLS coefficient variance under the classical homoskedastic model

$$\mathrm{Var}(\hat{\beta}_j \mid X) = \frac{\sigma^2}{\underbrace{\sum_i (x_{ij} - \bar{x}_j)^2}_{\mathrm{SST}_j} \quad \underbrace{(1 - R_j^2)}_{\text{independent variation left}}}$$

What is R_j^2?

The R^2 when X_j is regressed on the other predictors. A large value means most of its variation is shared.

What happens?

As $R_j^2 \to 1$, coefficient variance can rise; individual t-statistics and confidence intervals become less informative.

$\mathrm{VIF}_j = 1/(1 - R_j^2)$. Under the usual exogeneity assumptions, near multicollinearity does **not** itself create OLS coefficient bias.

Prediction: useful signals can survive coefficient instability

$$\widehat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$$

When $X_1 \approx X_2$ on the support of the data,

$$\widehat{Y} \approx \hat{\beta}_0 + (\hat{\beta}_1 + \hat{\beta}_2) X_1.$$

Economic intuition

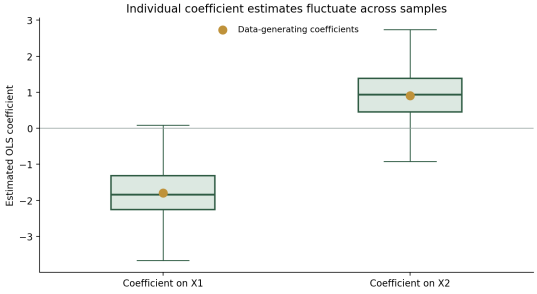
Individual effects may be difficult to disentangle, while their *combined forecast contribution* can be estimated with useful accuracy.

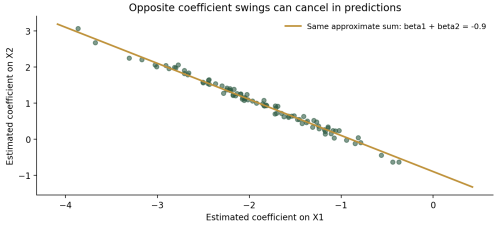
Model selection follows the task

If the question is *prediction*, evaluate the combined forecast on new observations using MAE and RMSE, rather than selecting solely by individual p-values.

The approximation applies when new observations follow the same correlated predictor relationship; it may fail outside that support.

Simulation: coefficients can move in opposite directions



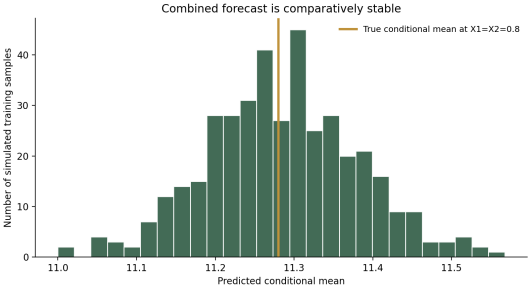


How to read these figures

Left: the boxes show OLS coefficient distributions across 400 simulated training samples; gold dots mark true coefficients. **Right:** when one coefficient rises, the other often falls, partly cancelling in their sum.

Simulated: $n = 80$ per replication, predictor correlation 0.995, outcome $Y = 12 - 1.8X_1 + 0.9X_2 + \varepsilon$, noise SD 0.65. Not Week 1 empirical results.

Simulation: the joint forecast is more stable



What was evaluated?

- Each of 400 samples generates a new OLS fit.
- Predict at $X_1 = X_2 = 0.8$.
- Gold line = the true conditional mean.

Read the comparison

Coefficient SDs are 0.76 each, while the SD of their *combined predicted mean* is only 0.096. This separates coefficient uncertainty from forecast stability.

Monte Carlo illustration at a point on the observed predictor support. Not a 2025 poverty forecast or test RMSE.

Does prediction simply ignore multicollinearity?

When it may matter less

- New observations resemble the training data.
- The shared predictor signal is strong.
- Holdout predictions remain accurate and stable.

When it may matter more

- Samples are small or coefficients highly variable.
- Correlation patterns shift over time.
- Predictions extrapolate beyond the training support.
- Tuning overfits a validation sample.

Appropriate predictive criterion

Do not automatically delete a correlated variable. Compare forecast errors and stability under a credible time-respecting validation design. More predictors are not necessarily better.

Regularization and correlated economic indicators

	OLS	Ridge	LASSO
Objective	Squared-error loss	Loss + L_2 penalty	Loss + L_1 penalty
Correlated X	Coefficients can offset one another	Shrinks coefficients and can stabilize predictions	Can select some variables, but correlated choices may be unstable
Prediction	Evaluate 2025 holdout RMSE/MAE	Same evaluation	Same evaluation
Interpretation	Partial conditional associations	Shrunken associations	Shrunken or zeroed associations

Important caveat

Ridge or LASSO has *no automatic* accuracy advantage. Elastic Net combines L_1/L_2 penalties and can be useful with groups of correlated predictors.

Coefficient selection is not causal identification. The chosen penalty strength should be validated using training-era observations only.

Optional R experiment: inspect rather than assume

Experiment A | Existing six predictors

1. Keep Week 1's 2021–2024 training and 2025 test split.
2. Compare OLS, Ridge and LASSO on the same six lagged X.
3. Report common holdout MAE and RMSE.

Experiment B | Separate stress test

1. Add a deliberately correlated predictor to *both* samples.
2. Inspect correlation, VIF and fitted coefficient changes.
3. Reassess 2025 prediction error without tuning on 2025.

What should students report?

Coefficient changes and their uncertainty; training vs validation vs test RMSE; prediction gaps for each model. Treat Experiment B as an illustration, not the main like-for-like benchmark.

The existing Week 2 R package implements Experiment A. Experiment B is an optional exercise concept, not an empirical result already generated by that script.

A decision rule for an applied economist

When the question is inferential

Focus on credible design, coefficient uncertainty, sensitivity to specification, and whether the predictor has sufficient independent variation.

When the question is predictive

Focus on the forecast target, the available information set, validation, prediction stability, and held-out MAE/RMSE.

Bridge to the next topic

If correlated predictors can make coefficient estimation unstable, can we improve *forecast robustness* by reducing coefficient magnitudes? **Ridge and LASSO address this through loss plus a penalty.**

What does regularization add to the objective?

$$\underbrace{\text{Training loss}}_{\text{fit to observations}} + \underbrace{\lambda \times \text{Penalty}}_{\text{limit coefficient size}}$$

- The **loss** measures how poorly the fitted model matches observed training outcomes.
- The **penalty** adds a cost for large coefficients.
- $\lambda \geq 0$ controls the penalty's strength.

The regularized training objective changes; final model evaluation still uses held-out predictive loss.

Training objective and evaluation loss are not identical

Inside model fitting

The algorithm chooses coefficients to minimize

$$\text{MSE}_{\text{train}} + \lambda P(\beta)$$

Here, the penalty influences learning from labeled examples.

Outside model fitting

We choose and evaluate candidate models by predicting outcomes not used for that fit.

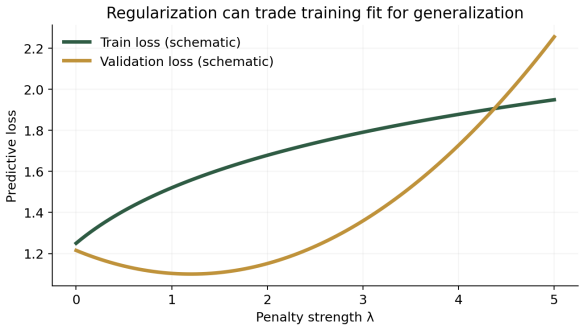
$$\text{RMSE}_{\text{validation}}$$

No coefficient penalty is added to reported forecast RMSE.

Example for students

A Ridge model may have a *higher* training MSE than OLS but a *lower* validation RMSE. The result is possible because controlling coefficient size can reduce sensitivity to training noise.

Figure reading: changing the penalty strength λ



Horizontal: λ (larger = stronger shrinkage).

Vertical: predictive loss.

Green: training loss may increase with penalization.

Gold: validation loss can be smallest at an intermediate λ.

Role in supervised ML

Use historical validation outcomes to choose λ. The value minimizing training loss is generally not the criterion for selecting predictive regularization.

Schematic curves only. In the final model the penalized training objective is minimized, while the plot shows unpenalized prediction loss.

Ridge regression: squared-coefficient penalty

$$\min_{\beta_0,\boldsymbol{\beta}} \frac{1}{2n}\sum_{i=1}^{n}(y_i - \beta_0 - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \frac{\lambda}{2}\sum_{j=1}^{p}\beta_j^2$$

- Ridge adds an L_2 penalty to the squared-error objective.
- Correlated predictors can produce more stable predictions under shrinkage.
- Typically shrinks nonzero coefficients toward zero without setting them exactly to zero.

Formulation follows the Gaussian `glmnet` objective. Intercept is unpenalized and predictor standardization matters.

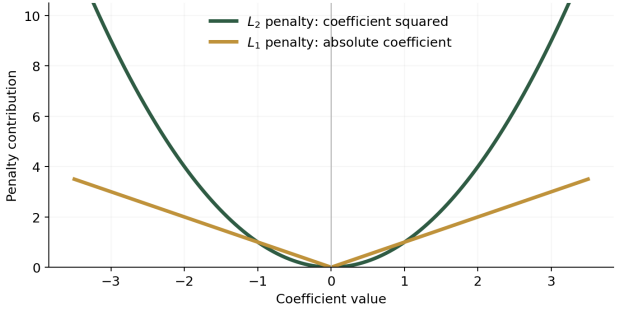
LASSO regression: absolute-coefficient penalty

$$\min_{\beta_0,\boldsymbol{\beta}} \frac{1}{2n}\sum_{i=1}^{n}(y_i - \beta_0 - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

- LASSO uses an L_1 penalty.
- Some coefficients can become exactly zero.
- Useful for prediction when selecting among candidate predictors may be beneficial.

Zero coefficients are algorithmic selections, not evidence that excluded variables have no economic importance or causal effect.

Figure reading: L_1 and L_2 penalize coefficients differently



X-axis: one coefficient β.

Y-axis: its penalty contribution.

Gold, L_1**:** $|\beta|$ has a sharp corner at zero.

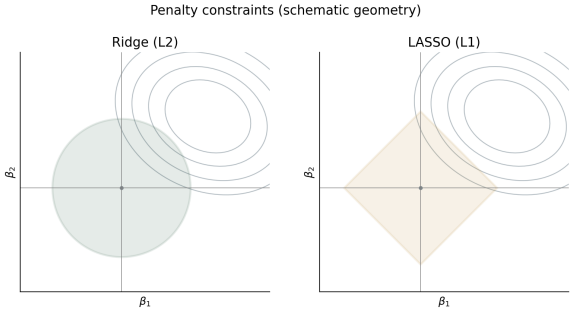
Green, L_2**:** β^2 is smooth at zero.

Role in supervised ML

The penalty works alongside the loss computed from observed Y. It influences the coefficients selected during training; L_1 can produce exact zeros.

These are mathematical penalty shapes, not measured predictive improvements.

Figure reading: why LASSO can select variables



Axes: two model coefficients, β_1 and β_2.

Contours: equal squared-error loss.

Ridge: round L_2 constraint.

LASSO: L_1 diamond with corners on the axes.

A solution at a corner can set a coefficient to zero.

Role in supervised ML

Regularization constrains coefficients while learning from labeled outcomes. Geometric corners explain LASSO sparsity, not causal variable importance.

Schematic constrained-regression representation, shown only for two predictors.

OLS, Ridge and LASSO: an undergraduate comparison

	OLS	Ridge	LASSO
Squared-error loss	Yes	Yes	Yes
Penalty	None	L_2	L_1
Requires λ tuning	No	Yes	Yes
Coefficients shrink	No explicit shrinkage	Yes	Yes
Exact zeros	Not typically from penalty	Generally no	Possible
Predictive advantage over OLS	Baseline	Not guaranteed	Not guaranteed
Economic interpretation	Conditional associations	Shrunken associations	Shrunken/selected associations

What stays constant in our experiment?

Outcome, six lagged economic predictors, final 2025 holdout and evaluation metric.

Elastic Net: know the idea, not a fourth full lab

$$\frac{1}{2n}\|y - \beta_0 - X\beta\|_2^2 + \lambda\left[\alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2\right]$$

$\alpha = 0$	$0 < \alpha < 1$	$\alpha = 1$
Ridge	Elastic Net	LASSO

In `glmnet`, α selects the penalty mix while λ sets penalty strength. Elastic Net is optional in this undergraduate workshop.

Why standardizing economic predictors matters

Different numerical units

- GRDP per capita: million IDR
- Unemployment: percent
- Schooling: years
- Internet access: percent

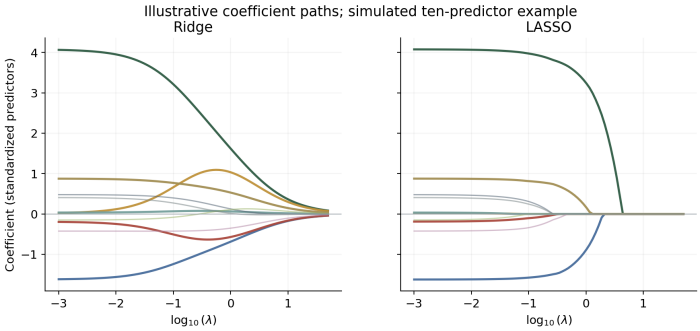
Penalty implication

- Coefficient sizes depend on predictor units.
- A common penalty can behave unevenly across raw scales.
- Standardize using training data only.

$$x_{ij}^{\text{std}} = \frac{x_{ij} - \overline{x}_{j,\text{train}}}{s_{j,\text{train}}}$$

The R `glmnet` option `standardize=TRUE` computes its scaling from the data passed to each fit. Do not use test-year statistics in preprocessing.

Figure reading: Ridge and LASSO coefficient paths



Horizontal: $\log_{10}(\lambda)$; move right for stronger penalty.

Vertical: fitted coefficient values.

Each curve: one predictor's coefficient.

Ridge: smooth shrinkage toward zero.

LASSO: some coefficients reach exactly zero.

Role in supervised ML

Coefficient paths reveal how the trained prediction rule changes as complexity control increases, *before* selecting λ using validation.

Illustrative 10-predictor simulation, not the six economic coefficients from Week 1.

What do coefficient paths mean for economic predictors?

Possible interpretation

- Schooling and internet access may contain related information.
- Ridge can retain both with smaller coefficients.
- LASSO may set one coefficient to zero at some λ.

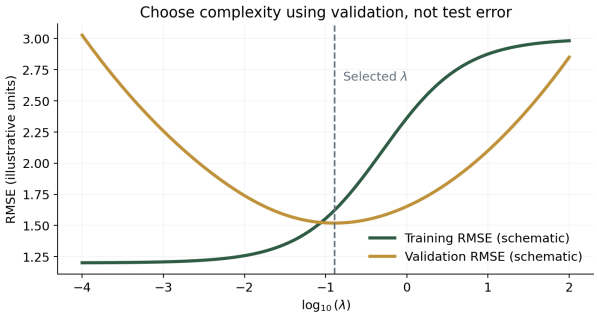
What not to conclude

- A zero LASSO coefficient does not imply that education has no economic role.
- A coefficient path does not prove a causal relationship.
- Selection can vary with correlated variables and samples.

Teaching task

Ask students to compare the number of nonzero coefficients at three λ values. Then ask which model should be preferred. **Answer: coefficient paths alone cannot decide; validation performance is needed.**

Figure reading: why we tune λ using validation



Horizontal: $\log_{10}(\lambda)$, increasing to the right.

Vertical: RMSE, where smaller is better.

Green: training RMSE; increasing penalty worsens training fit.

Gold: validation RMSE, with a minimum at an intermediate penalty.

Dashed line: selected λ from validation.

Role in supervised ML

Observed validation targets supply feedback for choosing the tuning parameter. The untouched 2025 poverty outcomes cannot be used to choose λ.

Schematic RMSE curves, not estimated Ridge/LASSO results.

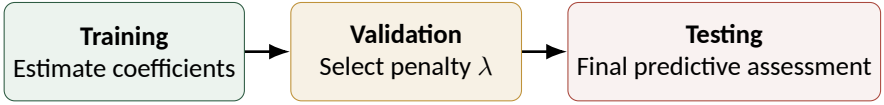
Putting the two figures together: paths and validation

	Coefficient paths	**Validation curve**
Horizontal dimension	Penalty strength λ	Penalty strength λ
Vertical dimension	Individual regression coefficients	Prediction error (RMSE)
What it tells us	How the fitted model changes	How well each candidate predicts unseen validation outcomes
What it cannot tell us alone	Which λ predicts best	Which variables have causal effects

Economic-prediction sequence

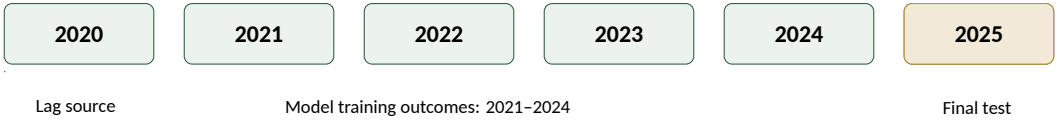
Fit candidate Ridge/LASSO models using past observed poverty $Y \rightarrow$ **inspect coefficient paths** $\rightarrow$ **choose** λ **by historical validation RMSE** $\rightarrow$ **refit** $\rightarrow$ **assess 2025 predictions once**.

Training, validation and test perform distinct tasks



- Fit coefficients using available training data.
- Choose the penalty from validation performance.
- Evaluate the selected model *once* on the untouched final holdout.

Preserve the chronological prediction design



Ex-ante prediction for 2025

Observed 2024 economic indicators may predict 2025 poverty. The 2025 poverty outcomes must remain unused during tuning and model choice.

Expanding-window validation: the preferred design

Validation fold	Fit with outcomes from	Predict and score outcome in
1	2021	2022
2	2021–2022	2023
3	2021–2023	2024
Final fit	2021–2024	**2025 final test**

Why not random row-level folds?

Observations within a year and region can be dependent. Forward-in-time validation better matches the operational question: predict a future year using earlier data.

References: Kuhn and Silge, *Tidy Modeling with R*, Chapter 10; time-respecting resampling.

Every validation fold must repeat preprocessing

1. Choose the fold's **historical fitting window** and later validation year.
2. Estimate missing-value medians **using only its fitting window**.
3. Apply those medians to the fold's validation records.
4. Let `glmnet` standardize using only the supplied fitting matrix.
5. Compute MAE/RMSE on that validation year.

No leakage

Do not preprocess all 2021–2024 observations once and then validate on 2022 or 2023. That would allow information from the validation year into preprocessing.

How is λ selected?

$$\widehat{\lambda} = \arg\min_{\lambda \in \Lambda} \frac{1}{K} \sum_{k=1}^{K} \mathsf{MSE}_k(\lambda)$$

- Train candidate models across a penalty grid Λ.
- Compute validation loss for each candidate penalty.
- Average the validation losses across historical folds.
- Select a penalty, refit with 2021–2024, and evaluate on 2025.

This formula equally weights the yearly folds; observation-weighted aggregation is another defensible choice. Keep the aggregation rule fixed before viewing 2025 errors.

What will the empirical comparison test?

Research contrast	What students should check
Multiple OLS vs Ridge	Can shrinkage reduce RMSE on 2025?
Multiple OLS vs LASSO	Does removing predictors improve accuracy?
Training vs test loss	Which model generalizes, not merely fits?
Coefficient values	Which estimates become smaller or zero?
Prediction errors by zone	Are errors concentrated among particular fictitious economic zones?

Do not promise a particular ranking

On the six-predictor Week 1 data, OLS may remain the best option. That result would be substantively useful for teaching.

Practical setup: continue directly from Week 1

Re-use the existing objects

`panel`: lagged predictors, not imputed

`train`: imputed outcomes 2021–2024

`test`: imputed outcomes 2025

`lag_predictors`: six feature names

`ols_model`: multiple OLS benchmark

New package

`glmnet` for Ridge and LASSO.

Run the Week 1 script first, or rebuild the same lag and split objects from the workbook.

The original data remain fully synthetic. Reproducing Week 1 OLS on the same sample allows a like-for-like comparison.

R demo 1: construct matrices for penalized regression

```r
# Run Week 1 first: train, test, lag_predictors and ols_model exist.
install.packages("glmnet")  # only if needed
library(glmnet)

x_train <- as.matrix(train[, lag_predictors])
x_test  <- as.matrix(test[, lag_predictors])
y_train <- train$poverty_rate_pct
y_test  <- test$poverty_rate_pct

stopifnot(ncol(x_train) == 6,
          nrow(x_train) == 480,
          nrow(x_test) == 120)
```

Instructor explanation

The outcome and predictors are identical to Week 1. The intercept is estimated internally and is not penalized by default.

R demo 2: estimate Ridge and LASSO paths

```r
lambda_grid <- 10^seq(-4, 2, length.out = 75)

ridge_path <- glmnet(x_train, y_train, alpha = 0,
                     lambda = lambda_grid, standardize = TRUE)
lasso_path <- glmnet(x_train, y_train, alpha = 1,
                     lambda = lambda_grid, standardize = TRUE)

plot(ridge_path, xvar = "lambda", main = "Ridge paths")
plot(lasso_path, xvar = "lambda", main = "LASSO paths")
```

Ask students

Which lines shrink, and do any LASSO coefficients become exactly zero? At this stage, no λ has been chosen by validation.

R demo 3: choose lambda without seeing 2025

Introductory route

- Sub-train on 2021–2023.
- Validate on 2024.
- Choose the penalty with the lowest 2024 loss.
- Refit on 2021–2024.

Recommended extension

- Repeat for validation years 2022, 2023, 2024.
- Use expanding training windows.
- Average across fold losses.
- Refit and evaluate once on 2025.

The single-year holdout within training is easiest to code live. The expanding-window approach is more stable as a model-selection exercise and is preferred for the full workshop script.

R demo 4: a clean 2024 validation split

```r
# panel contains the lagged variables BEFORE imputation.
subtrain <- subset(panel, year <= 2023)
valid    <- subset(panel, year == 2024)

# Use only subtrain to learn missing-data replacements.
med <- vapply(lag_predictors, function(v)
  median(subtrain[[v]], na.rm = TRUE), numeric(1))

for (v in lag_predictors) {
  subtrain[[v]][is.na(subtrain[[v]])] <- med[[v]]
  valid[[v]][is.na(valid[[v]])]       <- med[[v]]
}
```

The validation-year outcomes are not used for any imputation median; the 2025 holdout is not accessed here.

R demo 5: calculate validation RMSE along the path

```r
xx <- as.matrix(subtrain[, lag_predictors])
xv <- as.matrix(valid[, lag_predictors])
yy <- subtrain$poverty_rate_pct

tune_once <- function(alpha) {
  fit <- glmnet(xx, yy, alpha = alpha,
                lambda = lambda_grid, standardize = TRUE)
  predictions <- predict(fit, newx = xv, s = lambda_grid)
  squared <- sweep(predictions, 1, valid$poverty_rate_pct, "-")^2
  scores <- sqrt(colMeans(squared))
  lambda_grid[which.min(scores)]
}
```

A transparent first pass using a single 2024 validation year. Later, replace this by averaging expanding-window fold scores.

R demo 6: select lambda, refit and predict 2025

```
lam_ridge <- tune_once(alpha = 0)
lam_lasso <- tune_once(alpha = 1)

ridge_final <- glmnet(x_train, y_train, alpha = 0,
                      lambda = lambda_grid, standardize = TRUE)
lasso_final <- glmnet(x_train, y_train, alpha = 1,
                      lambda = lambda_grid, standardize = TRUE)

pred_ridge <- as.numeric(predict(ridge_final, x_test, s = lam_ridge))
pred_lasso <- as.numeric(predict(lasso_final, x_test, s = lam_lasso))
pred_ols <- as.numeric(predict(ols_model, newdata = test))
```

Now, and only now, evaluate the models on 2025. This code mirrors the Week 1 testing outcome.

R demo 7: compute the same error metrics as Week 1

```r
mae  <- function(y, p) mean(abs(y - p))
rmse <- function(y, p) sqrt(mean((y - p)^2))

comparison <- data.frame(
  model = c("Multiple OLS", "Ridge", "LASSO"),
  MAE  = c(mae(y_test, pred_ols),  mae(y_test, pred_ridge),
           mae(y_test, pred_lasso)),
  RMSE = c(rmse(y_test, pred_ols), rmse(y_test, pred_ridge),
           rmse(y_test, pred_lasso))
)
print(comparison)
```

Discussion prompt

Is the gain (if any) meaningful? Is it stable across validation folds? Would you choose a more complicated rule for a tiny difference?

Interpretation guide: what can results tell us?

Possible result	Interpretation
Ridge lower RMSE than OLS	Shrinkage improved this particular holdout prediction.
LASSO sets coefficients to zero	The penalty selected a sparser predictor set; it is not a causal finding.
OLS still has lowest RMSE	Six predictors and ample training data may make shrinkage less beneficial.
Different MAE / RMSE ranking	Some models may have especially large errors for a few observations.
Different zone-specific errors	Predictive reliability may vary across groups of fictitious regions.

These are decision rules for reading future output, not claims about uncomputed Ridge/LASSO results.

What the output table should look like

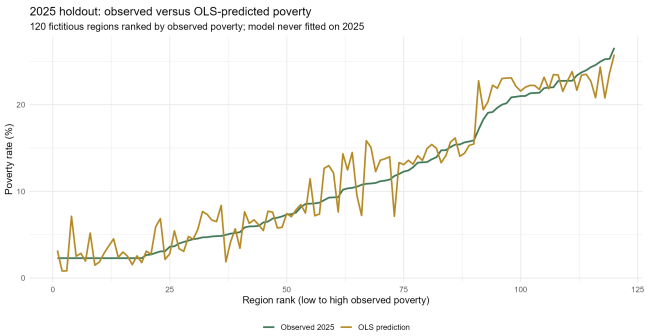
Model	Holdout MAE	Holdout RMSE	Penalty λ
Mean benchmark	6.957	7.827	–
Simple OLS	2.439	3.034	–
Multiple OLS	1.523	1.977	–
Ridge	*Run R*	*Run R*	*Validate*
LASSO	*Run R*	*Run R*	*Validate*

Validation rule

The reported 2025 errors are final comparison outcomes. Do not revise the chosen penalty or model based on this table and then report the same 2025 observations as an untouched test.

Week 1 numbers are from submitted outputs. Ridge and LASSO cells are intentionally unfilled pending R estimation.

Out-of-sample plot: continue the Week 1 diagnostic

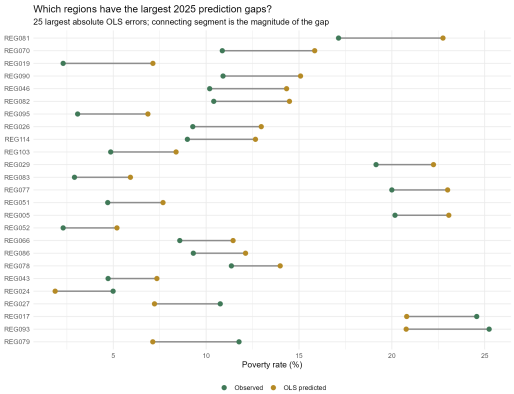


Extend in Week 2

- Keep observed 2025 poverty.
- Add Ridge predictions.
- Add LASSO predictions.
- Rank regions for display, not for estimating.

Shown graphic: actual Week 1 R output for Multiple OLS and observed 2025 outcomes only. New curves are to be generated from Week 2 estimation.

Prediction gaps: focus on where models make mistakes



Economic interpretation

- Which synthetic regions have the biggest OLS misses?
- Do Ridge / LASSO shrink the gaps?
- Are errors systematically positive or negative?
- Are poor predictions concentrated by economic zone?

Week 1 actual output: top 25 OLS absolute errors in the 2025 holdout. The plot is descriptive; it is not used to tune the models.

Expected output files for a Week 2 R demonstration

Deliverable	Teaching purpose
Loss and penalty illustrations	Connect optimization and shrinkage visually.
Ridge / LASSO paths	Inspect coefficients along λ.
Validation RMSE curve	Document parameter choice without viewing 2025.
Model comparison table	Compare OLS, Ridge, and LASSO on 2025.
Actual–predicted figure	Interpret out-of-sample fit.
Prediction gaps	Diagnose regional prediction errors.
XLSX workbook	Preserve per-region predictions and metrics.

The names and numerical values of newly generated outputs depend on the final Week 2 demonstration script.

A second experiment: when many predictors are available

Scenario A: original six predictors

- Direct continuity with Week 1
- Most transparent interpretation
- May show little gain from regularization

Scenario B: expanded design

- Add interactions and transformations
- Introduce deliberately redundant predictors
- Examine shrinkage and overfitting

Teaching discipline

Mark the expanded predictor set as an illustrative specification. Build transformations without using information unavailable at the forecast date. Keep 2025 out of model selection.

Three misunderstandings to avoid

1. **"OLS is non-ML; Ridge is ML."** Both can be supervised prediction methods; the training objective differs.
2. **"A penalty always increases accuracy."** It trades training fit for complexity control; test outcomes may improve or worsen.
3. **"LASSO reveals the true causal drivers."** Variable selection for predictive accuracy is not an identification strategy.

Class discussion and mini-exercises

1. What does changing from MSE to MAE do to the relative cost of large errors?
2. Why is λ selected with validation instead of the final test sample?
3. What happens to Ridge/LASSO coefficients when the penalty grows?
4. If Ridge RMSE is 1.98 and OLS RMSE is 1.977, which would you recommend, and why?
5. Does a LASSO zero coefficient mean an economic variable is unimportant?

Takeaways: the five links in predictive regression



Core idea

A useful ML regression workflow combines **explicit objectives**, **complexity control**, and **credible out-of-sample evaluation**. A particular method is preferred only when the evidence supports its predictive performance and practical usefulness.

Pattern Discovery in Economic Data

The Use of Unsupervised Machine Learning

From supervised prediction

- Observed outcome/label Y in training
- Optimize a prediction objective
- Evaluate on unseen outcomes

To unsupervised structure

- No predefined target outcome
- Summarize indicators with PCA
- Group observations with clustering

Selected teaching references

- James, G., Witten, D., Hastie, T., and Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed., especially Chapter 6 (Linear Model Selection and Regularization). `statlearning.com`.
- Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software*, 33(1), 1–22.
- Wooldridge, J. M. (2020). *Introductory Econometrics: A Modern Approach*, 7th ed., multiple regression and multicollinearity discussion.
- `glmnet` documentation and vignette: definition of the Gaussian objective, α, λ, and standardization.
- Kuhn, M. and Silge, J. *Tidy Modeling with R*, Chapter 10: time-respecting resampling and performance assessment. `tmwr.org`.

The workshop dataset and Week 1 regression output are entirely synthetic, intended for instruction rather than empirical claims about Indonesia.